Systems:

A system is viewed as an interconnection of operations that transforms an input signal into an output signal with properties different from input signal.

Let the operator H denote the action of a system. Then application of a continuous-time signal $x(t)$ to the input of the system yields the output signal

$$y(t) = H\{x(t)\}$$

For discrete case: $y[n] = H\{x[n]\}$

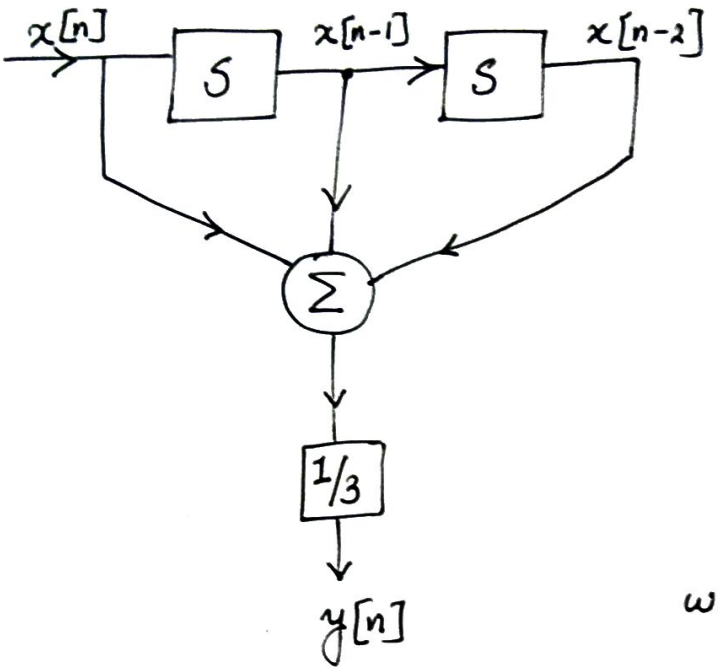
e.g. Matrix-vector multiplier etc.

Moving average system:

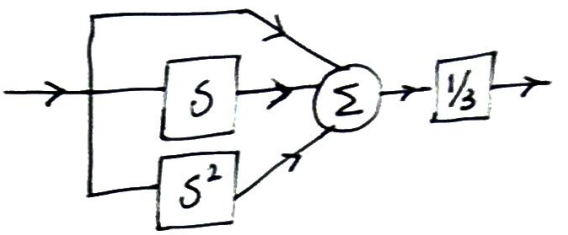
$$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$$

Q: Find the operator H for this system:

$$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$$



Cascade



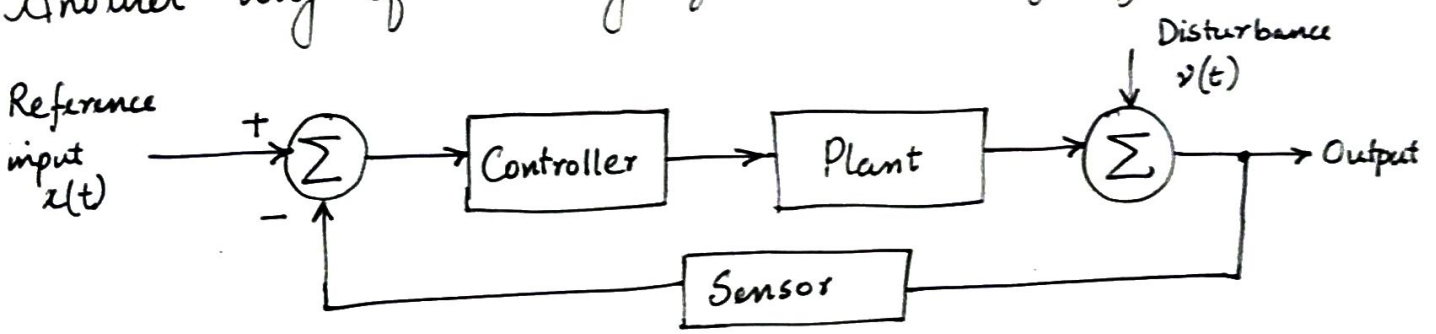
Parallel

$$H = \frac{1}{3}\left(1 + S + S^2\right)$$

where

In this example the signal flows in the forward direction.

Another way of combining systems is through feedback.



Properties of Systems:

1) Stability: A system is said to be bounded-input bounded-output (BIBO) stable if and only if every bounded input results in a bounded output.

$$y(t) = H\{x(t)\}$$

The operator H is BIBO stable if the output signal $y(t)$ satisfies the condition

$$|y(t)| \leq M_y < \infty \quad \forall\, t$$

whenever the input signals $x(t)$ satisfy the condition

$$|x(t)| \leq M_x < \infty \quad \forall\, t$$

Both M_x, M_y are some finite positive numbers.

- A system should remain stable under all possible operating conditions. Only then is a system BIBO stable.
- Unstable systems are generally avoided, unless some mechanism can be found to stabilize them.
- e.g. TACOMA NARROWS SUSPENSION BRIDGE (November 7, 1940 – 11 a.m).

Example : ① Show that

$$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$$

is stable.

Solution: Let $|x[n]| < M_x < \infty \ \forall n$

$$|y[n]| = \frac{1}{3}\left|x[n] + x[n-1] + x[n-2]\right|$$

$$\leq \frac{1}{3}\left(|x[n]| + |x[n-1]| + |x[n-2]|\right)$$

$$\leq \frac{1}{3}\left(M_x + M_x + M_x\right) = M_x$$

Hence, the absolute value of $y[n]$ is always less than the maximum absolute value of the input signal $x[n] \ \forall n$. Thus the MAS is stable.

② $y[n] = r^n x[n]$, where $r > 1$

Solution: $|x[n]| < M_x < \infty \ \forall n$ assume

$$|y[n]| = |r^n x[n]|$$

$$= |r^n| \cdot |x[n]| \quad (\text{Chose } x[n] = 1)$$

With $r > 1$, $|r^n|$ diverges with increasing n.

UNSTABLE SYSTEM

② Memory: A system is said to possess memory if its output signal depends on past or future values of the input signal.

Memoryless: Output signal depends only on the present value of the input signal.

— Resistor:

$$i(t) = \frac{1}{R} v(t)$$

Damping:

$$F = B v$$

velocity

force

Frictional co-efficient

— Inductor:

$$\underbrace{i(t) = \frac{1}{L} \int_{-\infty}^{t} v(\tau)\, d\tau}_{\text{Current through L depends on all past values.}}$$

$$F = K \int_{-\infty}^{t} v(\tau)\, d\tau$$

Spring constant

③ Causality: A system is said to be causal if the present value of the output signal depends only on the present or past values of the input signal.

Causal: $y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$

Noncausal: $y[n] = \frac{1}{3}\left(x[n+1] + x[n] + x[n-1]\right)$

Causality is required for a system to be capable of operating in real time.

④ Invertibility: A system is said to be invertible if the an unique input of the system can be recovered from the output.

Formally: Let $\mathcal{H}$ represent a continuous-time system, with the input signal $x(t)$ producing the output signal $y(t)$.

$$y(t) = \mathcal{H}\{x(t)\}$$

Let $\mathcal{G}$ represent the system that operates on $y(t)$.

$$\mathcal{G}\{y(t)\} = \mathcal{G}\mathcal{H}\{x(t)\} = x(t) \text{ if and only if } \mathcal{G}\mathcal{H} = \mathcal{I}.$$

e.g. $x(t) = \begin{bmatrix} 2 & 6 \\ 3 & 9 \end{bmatrix} y(t)$ $\qquad$ $x(t) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} y(t)$

$$x(t) = \begin{bmatrix} 2e^{-t} & 6e^{t} \\ 3e^{-t} & 9e^{t} \end{bmatrix} y(t) = \begin{bmatrix} 2 & 6 \\ 3 & 9 \end{bmatrix} \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{t} \end{bmatrix} y(t)$$

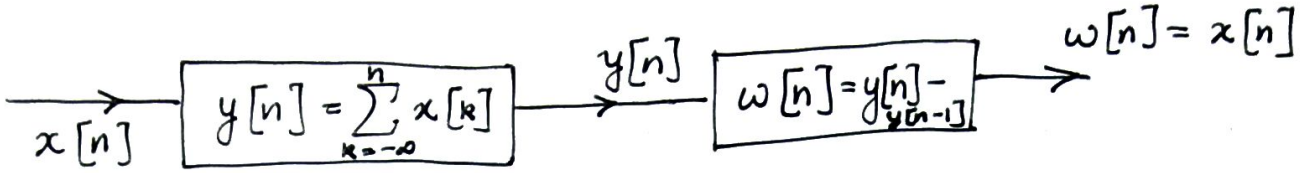
$y(t) = x^2(t)$

$\mathcal{G}$ is called the inverse operator of $\mathcal{H}$. The system corresponding to $\mathcal{G}$ is called the inverse system.

A system is invertible only if distinct inputs produces distinct output.

e.g. When a transmitted signal propagates through a communication channel, it gets distorted due to channel noise. An equalizer is used in receiver which acts as inverter to the channel distortion.

e.g. Accumulator : $y[n] = \sum_{k=-\infty}^{n} x[k]$



e.g. A signal is first applied as the input to a system known as an encoder (reason-encryption etc.). For lossless coding, the input to the encoder must be exactly recoverable from the output, i.e., the encoder must be invertible.

Time Invariance :

A system is time invariant if a time shift in the input signal results in an identical time shift in the output signal.

e.g in an RLC circuit, if R, L, C are constant over time then the system is time invariant. However, if R, L, C changes with time it is no longer time invariant.

e.g. $y(t) = \sin[x(t)]$

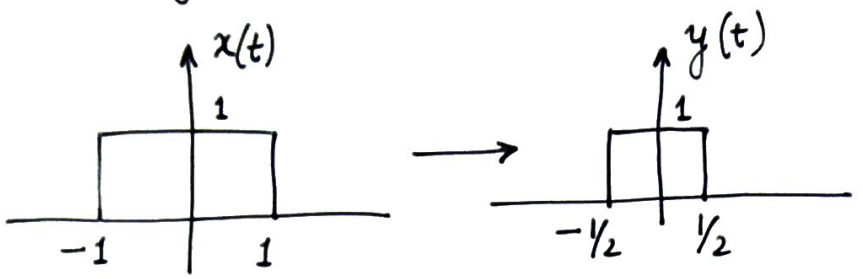
$y_1(t) = \sin[x_1(t)]$

$\sin[x_1(t-t_0)] = y(t-t_0) = y_1(t-t_0)$

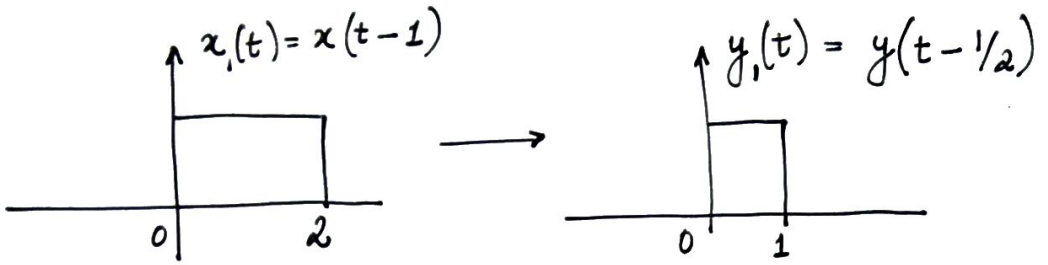
} Time-invariant

e.g. $y(t) = t\,x(t)$ — Time-variant

e.g. $y(t) = x(2t)$



$\Rightarrow$ Time variant



Linearity:

A linear system is a system that possesses the important property of superposition.

Let $y_1(t)$ and $y_2(t)$ be the output of a system corresponding to input $x_1(t)$ and $x_2(t)$. Then the system is linear if:

1. The response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$. (Additivity)
2. The response to $a x_1(t)$ is $a y_1(t)$. (Scaling)

Linear system:

$$a x_1(t) + b x_2(t) \longrightarrow a y_1(t) + b y_2(t)$$

$$a x_1[n] + b x_2[n] \longrightarrow a y_1[n] + b y_2[n]$$

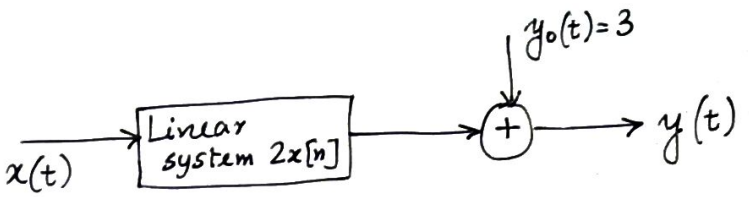
$\boxed{\square}$ An input which is zero for all time results in an output which is zero for all time.

e.g. $y(t) = t\,x(t)$ — Linear time-variant

$y(t) = x^2(t)$ — Nonlinear time-invariant

$y[n] = Re\{x[n]\}$ — Additive but not homogenous, time-invariant
(multiply by j)

$y[n] = 2x[n] + 3$ — Non-linear time invariant.



Q: $y[n] = x[n]\{g[n] + g[n-1]\}$

$g[n] = 1 \;\forall\; n \Rightarrow$ Time invariant

$g[n] = n \Rightarrow$ Time variant

$g[n] = 1 + (-1)^n \Rightarrow$ Time invariant.